

NutriFilter: VLM-Driven Dietary Augmentation in XR Passthrough Environments

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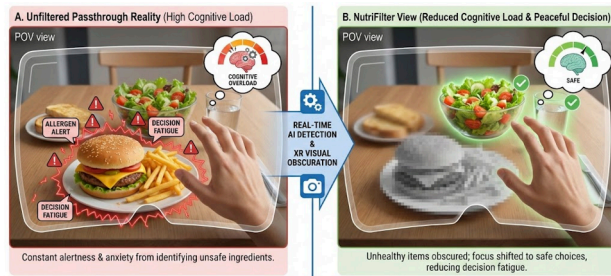


Figure 1: We propose an AI+XR system (NutriFilter) that supports users in correcting eating behavior. The figure visualizes the overview idea where our system changes A) an unfiltered passthrough scene to B) a NutriFilter-enabled scene.

ABSTRACT

Managing strict medical dietary restrictions imposes a severe cognitive load, which often leads to decision fatigue. We present NutriFilter, a novel mediated reality framework that modulates the visual saliency of restricted foods within a mixed reality environment. Our research explored the visual saliency options, like chromatic alteration, Gaussian blurring, and total occlusion. We developed a proof-of-concept prototype to demonstrate the feasibility of utilizing a vision-language model to enhance the food scene. By dynamically filtering visual stimuli to visually mask items, NutriFilter demonstrates the efficacy of context-aware Extended Reality interfaces in mitigating cognitive burden and supporting adherence to complex dietary protocols.

Index Terms: XR, Eating Behavior, AI, Diminished Reality.

1 INTRODUCTION

The act of eating becomes a major source of anxiety, as individuals must constantly check ingredients. Consuming a certain food item could be life-threatening or, for chronic conditions, a significant contributor to long-term health issues. Moreover, this examination of ingredients is not simple. It involves reading small-print nutrition labels with non-intuitive terminology or disguised ingredient names, confusing many ordinary people.

This task also becomes exponentially difficult in public environments where ingredients in sauces or preparations are often undisclosed and can be visually fine. However, the real problem is that choosing a safe meal is a constant, relentless process for every meal and food item. This state of constant alertness places a significant cognitive load on these individuals, exhausting the brain's limited processing capacity.

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When this cognitive load is sustained, it leads to decision fatigue, a state where a person's decision-making ability deteriorates. Just as a fatigued muscle, the brain's capacity for self-control is compromised after making numerous decisions about meals throughout the day. This fatigue can lead to several negative consequences, including impulsivity (such as consuming harmful food), decision avoidance (skipping meals), or errors in judgment (overlooking an important ingredient). Thus, when these dietary restrictions are broken, they can pose a risk to immediate health complications and long-term damage. The objectives of our work are as follows.

- XR Interface to facilitate healthy eating behavior by obscuring (e.g., blurring, hiding, coloring) the unhealthy food
- Explore and analyze the method of obscuring the food in the XR Interface
- Implications and design guidelines for facilitating an XR interface for healthy eating behavior

2 RELATED WORK

The classical approach to addressing the problem of consuming harmful meals has been to utilize Cognitive Behavioral Therapy (CBT) to modify eating patterns. This method helps patients persevere through distorted thoughts when they reach high levels of decision fatigue, allowing them to control their diets and the food items they consume [3]. However, some new applications have also been developed, allowing people to track their diet [4]. Nevertheless, one flaw that prevents these applications from being reliable sources is that users must input their data through self-reporting of daily meals. Users' reporting significantly decreases the accuracy of these records, as self-reported energy intake is often notoriously inaccurate compared to objective measurements [1]. To address this issue, our research will develop a proactive approach to this problem by creating an XR passthrough environment where users can take live pictures of the meal. Then, we will utilize an AI ecosystem to identify and blur harmful food items.

There are also new applications utilizing Augmented Reality (AR) to support dietary control in the real world. For instance, some systems designed for smart glasses can overlay nutritional data, such as calorie counts, directly onto physical food items [5]. This provides immediate information, which is significant in helping them resist temptation. This enables users to make informed decisions without having to read confusing nutritional labels.

As mentioned before, one flaw current technology has that prevents these applications from being reliable sources is that most remain reactive. They present data that relies entirely on the user's willpower to receive it, and the users are still the ones making the final choice. Knowing that a burger is high in calories does not remove the visual impulse to eat it, often failing when decision fatigue and temptation take control. To address this issue, our research will develop a perceptual intervention utilizing VR Passthrough and Diminished Reality techniques [2, 6], which will actively obscure harmful food items from the user's view, eliminating the visual stimulus entirely rather than just labeling it.

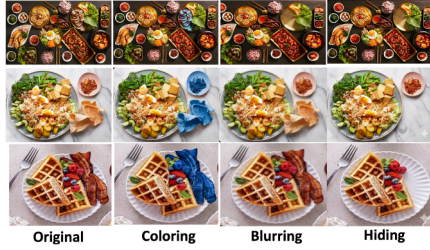


Figure 2: Potential augmented visualization methods for NutriFilter.

3 NUTRIFILTER METHODOLOGY

To explore the design space of the NutriFilter visualization method, we implemented several methods as shown in Figure 2. We explored three distinct Augmented Reality (AR) visualization methods as follows.

1. **Chromatic Alteration (“Coloring”)** Artificially recoloring the harmful food to an unnatural or alerting hue (e.g., turning pork blue or red) to signal danger or reduce appetite.
2. **Gaussian Blur (“Blurring”)** Applying a heavy blur filter to the specific region of interest (ROI) to reduce visual saliency.
3. **Diminished Reality (“Hiding”)** Digitally removing the object from the scene entirely (inpainting/masking) so it appears to be absent.

3.1 System Implementation

We developed a proof-of-concept prototype for the proposed AI+XR system (Figure 3). Here, we designed to reduce decision fatigue by proactively obscuring unhealthy food items in a user’s field of view.

The overall system follows a three-stage pipeline beginning with **Input & Capture**, where a Meta Quest 3 headset utilizes its Passthrough API to capture the real-world environment and stream the video feed to a host PC. During the **Processing & Analysis stage**, this video stream is managed within a Unity application that splits the data into two paths: the continuous raw video is routed to the XR Compositor to serve as the background reality, while specific frames (activated by the controller button) are captured and transmitted to the cloud-based GPT-4o Vision Language Model (VLM) for semantic analysis. The AI identifies harmful food items based on dietary restrictions and creates an image that conceals them. Finally, in the **Output & Intervention stage**, these generated images are fed into the visual overlay in the XR scene to create a “Diminished Reality” experience, where unhealthy foods are hidden while safe options remain visible to the user.

The proposed system, NutriFilter, was successfully implemented to demonstrate the end-to-end workflow of detecting and hiding specific food items in a Mixed Reality (MR) environment, as shown in Figure 4. When the user captures the real-time passthrough scene with the trigger button, the system passes the image to the VLM to detect unhealthy food. Then, the system creates an augmented overlay to obscure the identified item.

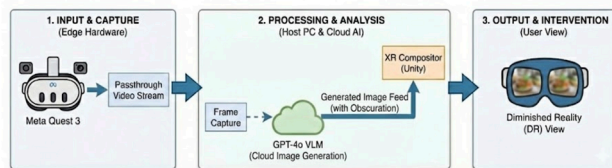


Figure 3: NutriFilter System Overview.



Figure 4: (A) A user clicks the button to activate the NutriFilter. (B) The integrated GPT-4o VLM detects harmful food (e.g., pork) and fully hides the unhealthy food from the real-time passthrough scene.

4 FUTURE WORK & CONCLUSION

Future development will focus on evolving the current proof-of-concept into a robust, real-time application by developing 3D spatial mapping and mask generation techniques to accurately obscure physical food items in 3D space. Currently, there is an additional hurdle of creating a synthetic image, which introduces an inherent delay for visual augmentation. Thus, employing real-time visualization will strengthen the advantage of NutriFilter.

In this work, we address the critical challenge of decision fatigue faced by individuals with strict dietary restrictions by introducing NutriFilter, a proactive augmented visual filter for the XR passthrough scene. NutriFilter utilizes the “Diminished Reality” concept, where we visually remove harmful food items from the user’s view before they trigger a temptation. We successfully implemented a proof-of-concept prototype integrating the Meta Quest 3 with GPT-4o VLM to detect and create the same scene image without harmful food items such as pork. We believe the system has the potential to reduce the cognitive load required to identify safe meals, thereby transforming the dining experience from a source of anxiety into a manageable process.

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